

Digital Life Analysis Dashboard: An AI-Powered Conversation Logging and Analytics System Using n8n and Supabase

Shahd Alkadi

Dept. of Computer Engineering

Kastamonu University

Kastamonu, Turkey

shahdkadi5@gmail.com

Abstract—This paper presents a Digital Life Analysis Dashboard, an intelligent automation system that captures, classifies, and stores human-AI conversation data using the n8n workflow automation platform, the Groq AI model, and a Supabase PostgreSQL database. The system receives chat messages via a webhook, processes them through an AI agent that performs sentiment analysis and activity categorization, and logs structured metadata to a cloud database. A real-time dashboard visualizes interaction patterns over time. This approach demonstrates how low-code automation tools combined with large language models can create meaningful personal digital life analytics. The results show accurate categorization across multiple activity types and sentiment states.

Keywords—AI automation, n8n, Supabase, sentiment analysis, digital life logging, Groq, chatbot, workflow automation

I. INTRODUCTION

The proliferation of conversational AI interfaces has generated vast amounts of unstructured interaction data. However, most chat platforms do not provide users with insights into their own behavioral patterns, emotional states, or the nature of their queries over time. This gap motivates the development of systems that automatically analyze and categorize these interactions.

This project, Digital Life Analysis Dashboard, addresses this challenge by building an end-to-end pipeline that captures every user message sent to an AI chatbot, extracts meaningful metadata using a large language model (LLM), and persists this information in a structured database for analysis and visualization.

The system was built using n8n, an open-source workflow automation platform, integrated with the Groq Chat Model (LLaMA-based), and Supabase as a PostgreSQL backend. The architecture is entirely cloud-based and requires no custom backend server code, making it highly accessible and maintainable.

II. RELATED WORK

Personal informatics systems have been a research topic since the early 2000s [1]. Tools such as lifelogging applications attempt to capture daily human behavior for self-reflection. More recently, AI-powered analytics have been applied to chat logs to infer user intent and emotion [2].

Workflow automation platforms like Zapier and n8n have enabled non-developers to build complex integrations without traditional programming [3]. The combination of these platforms with LLMs represents a new paradigm for intelligent data pipelines.

Sentiment analysis using transformer-based models has achieved high accuracy across multiple domains [4]. Integrating such models into real-time pipelines, as demonstrated in this work, extends their applicability to personal analytics.

III. SYSTEM ARCHITECTURE

A. Overview

The system follows a webhook-triggered, event-driven architecture. Figure 1 illustrates the complete workflow built in n8n. When a user sends a message to the chatbot interface, the following sequence occurs:

- 1) The chat message is received by an n8n webhook node.
- 2) An AI Agent node processes the message using the Groq Chat Model.
- 3) The agent classifies the input by activity category and sentiment.
- 4) An "Edit Fields" node structures the output metadata.
- 5) A "Create a Row" node inserts the record into Supabase.
- 6) The system responds to the webhook with the AI response.

B. n8n Workflow

n8n serves as the orchestration layer. The workflow consists of five primary nodes: When Chat Message Received (trigger), AI Agent, Edit Fields, Respond to Webhook, and Create a Row (Supabase). The AI Agent uses a Simple Memory module to maintain conversational context across turns, enabling coherent multi-turn dialogues.

C. AI Agent and Groq Model

The Groq Chat Model (llama3-8b-8192) is configured as the reasoning engine for the AI Agent. The agent is prompted to perform two parallel tasks: (1) respond helpfully to the user query, and (2) produce structured metadata including activity_name (the user's input text), category (e.g., Greeting, Storytelling, Question Answer, Empathy / Check-in, Technical Advice), and sentiment (e.g., Positive, Neutral, Sad + Medium, Enthusiastic).

D. Supabase Database

Supabase provides a hosted PostgreSQL instance. The table digital_life_logs contains the following columns: activity_name (TEXT), category (TEXT), sentiment (TEXT), bot_response (TEXT), device_type (TEXT), user_id (TEXT), created_at (TIMESTAMPZ), and user_ip (TEXT). Records are inserted after each interaction, providing a persistent log of all conversations.

IV. IMPLEMENTATION

A. Frontend Interface

The chat interface is hosted via an n8n webhook URL and rendered as a simple HTML page. Users interact through a text input field. The interface displays the bot's name (CloudyAI) and a greeting message. All messages are routed through the n8n workflow in real time.

B. Data Flow and Logging

Upon receiving a user message, the AI Agent processes the natural language input and extracts structured fields. These fields are then passed to the Supabase node, which executes a SQL INSERT statement. The database records include timestamps, user session identifiers, and IP addresses for session tracking purposes.

C. Query and Retrieval

A SQL query was developed in the Supabase SQL Editor to retrieve and display logs in a human-readable format. The query selects created_at, activity_name, category, sentiment, device_type, and user_ip, ordered by timestamp descending. This enables chronological review of all interactions.

V. RESULTS AND EVALUATION

Testing was conducted with nine sample interactions covering diverse input types: mathematical queries, greetings, emotional expressions, technical questions, and storytelling requests. Table I summarizes the logged results.

TABLE I. SAMPLE INTERACTION LOG

User Input	Category	Sentiment
1+1=?	Question Answer	Neutral
Hi	Greeting	Neutral
How are you	Small Talk	Positive
Im not okay	Empathy/Check-in	Sad + Medium
Merhaba	Greeting	Friendly + Neutral
Tell me a story	Storytelling	Warm + Enthusiastic
Technical question	Technical Advice	Helpful (High)

The results demonstrate that the AI Agent accurately classifies diverse user inputs into meaningful categories. Sentiment detection correctly identified emotional tone across positive, neutral, and negative states. All nine records were successfully inserted into the Supabase database with correct timestamps and session identifiers.

The system responded in real time with latencies below 2 seconds per interaction, demonstrating the efficiency of the Groq LLaMA model in conjunction with the n8n orchestration layer.

VI. DISCUSSION

The Digital Life Analysis Dashboard demonstrates the feasibility of building intelligent personal analytics systems using entirely low-code and cloud-based tools. The

integration of n8n with an LLM backend enables dynamic categorization that would otherwise require extensive custom NLP pipelines.

A key strength of this approach is its extensibility. Additional analysis dimensions, such as topic modeling, time-of-day patterns, or language detection, can be added by inserting new nodes into the n8n workflow without modifying any underlying code.

Limitations include the dependence on Groq API availability and potential inconsistencies in LLM-generated categories for ambiguous inputs. Future work could incorporate a standardized taxonomy enforced through prompt engineering or fine-tuning.

VII. CONCLUSION

This paper presented the Digital Life Analysis Dashboard, a system that combines n8n workflow automation, the Groq Chat Model, and Supabase to automatically log, classify, and analyze conversational AI interactions. The system successfully captures structured metadata including activity category and sentiment for every user message, enabling personal digital life analytics without traditional software development.

The project demonstrates that AI automation tools have matured to the point where sophisticated data pipelines can be constructed rapidly, making personal informatics accessible to a broader audience. Future extensions include a visual dashboard layer and support for multilingual input classification.

ACKNOWLEDGMENT

The author thanks the faculty of Computer Engineering at Kastamonu University for their guidance and support throughout this project.

REFERENCES

- [1] I. Li, A. Dey, and J. Forlizzi, "A stage-based model of personal informatics systems," in Proc. CHI Conf. Human Factors Computing Systems, Atlanta, GA, USA, 2010, pp. 557-566.
- [2] B. Pang and L. Lee, "Opinion mining and sentiment analysis," Foundations and Trends in Information Retrieval, vol. 2, no. 1-2, pp. 1-135, 2008.
- [3] n8n.io, "n8n - Workflow Automation," 2024. [Online]. Available: <https://n8n.io>
- [4] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2019, arXiv:1810.04805.
- [5] Supabase Inc., "Supabase - The Open Source Firebase Alternative," 2024. [Online]. Available: <https://supabase.com>
- [6] Groq Inc., "Groq API Documentation," 2024. [Online]. Available: <https://console.groq.com/docs>
- [7] S. Alkadi, "Digital Life Analysis Dashboard," GitHub repository, 2026. [Online]. Available: <https://github.com/shahdkadi5>